

SYMBOLS

Symbol	Means
Σ	"add them all up"
N / n	how many values: N = population, n = sample
$\mu / \bar{x}$	mean (average): μ population, $\bar{x}$ sample
σ / s	standard deviation: σ population, s sample
σ^2 / s^2	variance = $(SD)^2$
M, Q_1, Q_2, Q_3	median; quartiles at 25% / 50% / 75% ($Q_2 = M$)
IQR, LF, UF	$Q_3 - Q_1$: lower / upper fence (outlier cutoffs)
z	z-score = # of SDs from the mean
RF	relative frequency = part ÷ total (decimal)
$p / \hat{p}$	proportion: population / sample

Greek letter or capital N → POPULATION (everyone). Plain letter or small n → SAMPLE (some).

Every number answer: formula → plug in → arithmetic → boxed answer + units. RF as decimal (0.3000). Round probabilities to 4 decimals.

CH 1 — DATA COLLECTION

Population / Sample / Parameter / Statistic (Q1)

- **Population** = the **WHOLE** group you want to know about
- **Sample** = the smaller group actually measured (look for the number)
- **Individual** = ONE member
- **Parameter** = number describing the population — usually unknown
- **Statistic** = number computed from the sample — the number they give you

P-arameter → P-opulation S-tatistic → S-ample

Ex (Q1): diet study, 80 people, average drop 15 pts.
Population = all people who could use the diet · Sample = the 80 · Individual = one participant · Parameter = true mean drop for everyone (μ , unknown) · Statistic = 15 points ($\bar{x}$)

Process: 1 state the question → 2 collect data → 3 describe it → 4 inference (sample → population).

Variable types (Q2) — answer all 3

1. Qualitative or Quantitative?

- **Qualitative** = words / categories / labels. Includes rankings (1st, 2nd) and ID-type numbers (ZIP).
- **Quantitative** = numbers you can average.

2. Discrete or Continuous? (qualitative → write "Not applicable")

- **Discrete** = COUNTED, whole numbers ("number of...")
- **Continuous** = MEASURED, decimals possible (time, temp, weight, height, \$, age, %)

3. Level — ask in order:

1. Meaningful order? **NO** → **Nominal** (car type, brand, yes/no)
2. Equal gaps between values? **NO** → **Ordinal** (rankings, S/M/L, agree...disagree, letter grade, education level)
3. Does 0 mean "none"? **NO** → **Interval** (temperature °F/°C, year)
4. 0 = none? **YES** → **Ratio** (counts, \$, time, weight, height, age)

Qualitative → Nominal or Ordinal.
Quantitative → Ratio, except temperature / year = Interval.

Variable	Q/Q	D/C	Level
Car transmission (Q2a)	Qual	N/A	Nominal
Number of pets (Q2b)	Quant	Discrete	Ratio
Temperature °F (Q2c)	Quant	Continuous	Interval
Ranking top 5 players (Q2d)	Qual	N/A	Ordinal
Education level; Likert; young/middle/old	Qual	N/A	Ordinal
Phone type; provider; yes/no; ID	Qual	N/A	Nominal
Income \$; age (yrs); hours; grams; grade %	Quant	Continuous	Ratio
# students; # vending machines	Quant	Discrete	Ratio

Observational vs Experiment (Q3)

- **Explanatory variable** = the possible cause (input); **Response** = the outcome measured
- **Observational** = only watch / survey / record / track
- **Experiment** = researcher **ASSIGNS** the treatment (randomly assigned, given drug/placebo, control group)

Observational → association only → causation = NO.
Experiment → causation = YES.

- **Lurking variable** = not measured, but may be the real cause
- **Census** = measure everyone · **SRS** = every group of n equally likely: number everyone 1– N , generate n random numbers

Ex (Q3): survey, screen time vs GPA → Observational · Explanatory = screen time · Response = GPA · Causation? No.
Kids randomly assigned to music groups → Experiment. Teens tracked for years → Observational.

Bias (Q4) — ask in order

1. **Who got asked?** wrong/partial group → **Sampling bias** (only HQ, only readers, mall, online poll)
2. **Did they answer?** many didn't → **Nonresponse bias** (only 20% replied, refused, hung up)
3. **Are answers true?** no → **Response bias** (safe answers, leading question, lying)

Ex (Q4): sent only to headquarters → Sampling · 20% respond, mostly unhappy → Nonresponse · "safe" answers → Response

Sampling error = normal sample-vs-population difference. **Nonsampling error** = the biases above.

CH 2 — TABLES & GRAPHS

Category tables (Q5)

RF = count ÷ total · Degrees = RF × 360 · % = RF × 100
count = RF × total · missing RF = 1 – others

Checks: counts add to total · RFs add to 1 · degrees add to 360.

Ex (Q5, total 200): Sports 60 → $60/200 = 0.3000$ → $0.3000 \times 360 = 108^\circ$ · Movies 50 → 0.2500 → 90° · Shopping 40 → 0.2000 → 72° · Reading 30 → 0.1500 → 54° · Other 20 → 0.1000 → 36° .
Most popular = biggest count (Sports). Reading = 15%. Sports or Movies = $110/200 = 55\% > 50\%$ → True.

- **Bar graph** = one bar per category, gaps between. **Axis starts at 0, equal widths, 2-D.** Pareto = tallest → shortest.
- **Pie** = slices, angle = $RF \times 360^\circ$; parts of a whole. Bar to compare categories.
- Comparing two groups of different sizes → **use RF, not counts.**

Classes & histograms (Q6)

- **Lower / upper class limit** = first / last number of the class (60–69 → 60, 69)
- **Class width** = start of next class – start of this class (50 – 40 = 10, NOT 49 – 40)
- **Histogram** = bars for classes, touching. Most frequent class = biggest count (answer with the interval).
- How many in a range → add counts. Percent → counts ÷ total × 100.

Ex (Q6, 40 students; 40–49:4, 50–59:6, 60–69:10, 70–79:12, 80–89:8): 3rd class 60–69 → limits 60, 69 · width $50 - 40 = 10$ · ≥ 70 : $12 + 8 = 20$ → $20/40 = 50\%$ · < 60 : $4 + 6 = 10$ · biggest count 12 → 70–79

Shape (Q7) — look where the TAIL points

- **Bell-shaped** = hump in middle, symmetric: heights, weights, IQ
- **Uniform** = flat: fair die, random digits
- **Skewed RIGHT** = tail right, a few very BIG values: salaries, house prices, waiting times
- **Skewed LEFT** = tail left, a few very SMALL values: easy exam scores

Skewed right → mean > median · Skewed left → mean < median · Symmetric → mean ≈ median

Ex (Q7): salaries → skewed right · heights of boys → bell · die rolls → uniform · waiting times → skewed right

- **Stem-and-leaf** = stem = all digits but last, leaf = last digit (56 → 5 | 6); list every stem, leaves in order, add key.
- **Time-series** = time on bottom, value on side, dots connected.
- **Misleading graph** = axis not starting at 0, unequal bar widths, 3-D → fix: start at 0, equal widths.

CH 3 — NUMBERS THAT DESCRIBE DATA

Running example (Q8–Q11): commute times of ALL 10 employees (= population), sorted: 18, 22, 25, 28, 30, 31, 33, 35, 36, 40 N = 10

Mean (Q8a, Q9a)

Average: add all values, divide by how many.

$$\mu = \Sigma x \div N \text{ (population)} \cdot \bar{x} = \Sigma x \div n \text{ (sample)}$$

18+22+25+28+30+31+33+35+36+40 = 298 → $\mu = 298/10 = 29.8 \text{ min}$
Sample 18, 22, 25, 28, 30 → 123/5 → $\bar{x} = 24.6 \text{ min}$

Median (Q8c)

Middle value of SORTED data.

- Sort.
- Count n.
- n odd → middle value. n even → average the two middle values.

n = 10 (even) → 5th & 6th = 30, 31 → M = (30+31)/2 = 30.5 min
Odd: 76, 80, 84, 86, 90, 92, 94 → M = 86

Mode (Q8d)

Value that appears most. Two tie → both. Nothing repeats → "no mode".

Commute data: nothing repeats → no mode.
7, 12, 12, 20, 20, 25 → modes 12 and 20.

Shape from mean vs median (Q8e)

mean > median → skewed right · mean < median → skewed left · equal → symmetric

29.8 < 30.5 → skewed left (slightly)

Resistant = not thrown off by extreme values: median & IQR yes; mean, range, SD no. Symmetric → report mean & SD. Skewed → median & IQR.

Range

$$\text{Range} = \text{max} - \text{min}$$

40 – 18 = 22 min

Variance & Standard Deviation (Q8b population, Q9b sample)

SD = how far values typically are from the mean.
Variance = SD².

Population: $\sigma^2 = \Sigma(x - \mu)^2 \div N \quad \sigma = \sqrt{\sigma^2}$
Sample: $s^2 = \Sigma(x - \bar{x})^2 \div (n - 1) \quad s = \sqrt{s^2}$
"population"/all → ÷ N · "sample" → ÷ (n – 1)

- Find the mean.
- Column: value – mean.
- Column: square it.
- Add the squares.
- ÷ N (population) or ÷ (n – 1) (sample) → variance.
- √ → standard deviation.

x	x – μ	(x – μ) ²
18	–11.8	139.24
22	–7.8	60.84
25	–4.8	23.04
28	–1.8	3.24
30	0.2	0.04
31	1.2	1.44
33	3.2	10.24
35	5.2	27.04
36	6.2	38.44
40	10.2	104.04
add	0 ✓	407.60

Q8b: $\sigma^2 = 407.60/10 = 40.76 \text{ min}^2 \cdot \sigma = \sqrt{40.76} = 6.38 \text{ min}$
Q9b (sample 18, 22, 25, 28, 30; $\bar{x} = 24.6$): squares 43.56 + 6.76 + 0.16 + 11.56 + 29.16 = 91.20 → $s^2 = 91.20/4 = 22.8 \text{ min}^2 (\div 4, \text{ not } \div 5) \cdot s = \sqrt{22.8} = 4.77$

Calculator (1-Var Stats): σx = population SD, Sx = sample SD. Variance = square the UNROUNDED SD. Still write the table.

Empirical Rule (Q11) — bell-shaped only

68% within 1 SD · 95% within 2 SD · 99.7% within 3 SD
pieces: 34% | 13.5% | 2.35% | 0.15% (each side) · above mean = 50%

- Write μ and σ (round σ to 1 decimal if his numbers match that).
- Landmarks: $\mu \pm 1\sigma, \mu \pm 2\sigma, \mu \pm 3\sigma$.
- Find the question's numbers among the landmarks.
- Add the pieces between them.

10.6	17.0	23.4	29.8	36.2	42.6	49.0
2.35%	13.5%	34%	34%	13.5%	2.35%	

Q11 ($\mu = 29.8, \sigma = 6.4$): (a) 95% → $29.8 \pm 2(6.4) = 29.8 \pm 12.8 \rightarrow 17.0 \text{ to } 42.6$ · (b) 23.4 to 36.2 = $\pm 1 \text{ SD} \rightarrow 68\%$ · (c) 36.2 to 42.6 = 1 to 2 SD above → $(95 - 68)/2 = 13.5\%$
Quick: above $\mu + 1\sigma = 16\%$ · above $\mu + 2\sigma = 2.5\%$ · $\mu + 1\sigma$ to $\mu + 3\sigma = 15.85\%$

z-score & percentile

$$z = (\text{value} - \text{mean}) \div \text{SD} \text{ (bigger } z = \text{relatively higher)}$$

ACT 27 (mean 18, SD 6): $z = 9/6 = 1.50 \cdot \text{SAT}$
1200 (mean 1000, SD 200): $z = 1.00 \rightarrow \text{ACT student did better}$

95th percentile = 95% of data at or below. Median = 50th percentile.

Quartiles & five-number summary (Q10a)

- Sort.
- Q_2 = median of everything.
- Q_1 = median of lower half · Q_3 = median of upper half · n odd → leave the median OUT of both halves.
- Five-number summary = Min, Q_1 , M, Q_3 , Max

Lower 18, 22, 25, 28, 30 → $Q_1 = 25$ · Upper 31, 33, 35, 36, 40 → $Q_3 = 35$
18, 25, 30.5, 35, 40
Odd: 76, 80, 84 | 86 | 90, 92, 94 → $Q_1 = 80, Q_3 = 92$
32, 76, 80, 84 | 86, 90, 92, 94 → $Q_1 = 78, M = 85, Q_3 = 91$

IQR, fences, outliers (Q10b, c)

$$\text{IQR} = Q_3 - Q_1 \cdot \text{LF} = Q_1 - 1.5 \cdot \text{IQR} \cdot \text{UF} = Q_3 + 1.5 \cdot \text{IQR}$$

outlier = below LF or above UF

$\text{IQR} = 35 - 25 = 10 \cdot 1.5 \times 10 = 15 \cdot \text{LF} = 25 - 15 = 10 \cdot \text{UF} = 35 + 15 = 50 \cdot 18 > 10, 40 < 50 \rightarrow \text{no outliers}$
With outlier: $Q_1 = 78, Q_3 = 91, \text{IQR} = 13, 1.5 \times 13 = 19.5 \rightarrow \text{LF} = 58.5, \text{UF} = 110.5 \rightarrow 32 \text{ is an outlier}$

Boxplot

- Five-number summary + fences.
- Box from Q_1 to Q_3 , line at median.
- Whiskers to smallest / largest values INSIDE the fences.
- Outliers = separate *.



Read shape: median centered + equal whiskers → symmetric · median left + long right whisker → skewed right.

DON'T LOSE POINTS

- Sort before median / quartiles.
- "Sample" → ÷ (n – 1). "Population" → ÷ N.
- SD asked → take the √. Write units (variance = min²).
- σx = population, Sx = sample. Never square a rounded SD.
- Odd n → median stays OUT of the halves.
- Class width = 50 – 40 = 10, not 49 – 40.
- RF as decimal (0.3000). Degrees need °.
- Observational → NO causation.
- Skew = direction of the TAIL.
- "No mode" / "Not applicable" are real answers.
- Show every step.